

REPORT
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Carried out on

**“Temperature Controlled Child Care
System Using Arduino UNO”**

Submitted to

NMAM INSTITUTE OF TECHNOLOGY, NITTE

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In partial fulfillment of the requirements for the award of the

Degree of Bachelor of Technology

In

Electronics and Communication Engineering

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ABSTRACT

This project presents a Temperature Controlled Child Care System Using Arduino UNO, specifically designed for children aged 1 to 3 years. During the summer season, ambient temperature often rises beyond comfortable limits, which can disturb children's sleep and overall comfort. According to pediatric doctors, a surrounding temperature of around 28°C is considered comfortable and safe for children in this age group. When the temperature exceeds this limit, children may feel discomfort, wake up suddenly, and start crying or showing irritation.

In this system, a temperature sensor continuously monitors the surrounding temperature near the child. The sensed data is processed by the Arduino UNO, which compares it with the predefined threshold value of 28°C. When the temperature crosses this limit, an audio module connected to a speaker is activated to play a soothing or attention-based song. This sound indicates that the environment has become uncomfortable and immediate care is required. When the temperature remains within the safe range, the system remains silent. The proposed system is simple, low-cost, and effective, making it suitable for basic child care support during hot weather conditions, especially in summer.

PROJECT DETAILS

Introduction

Children aged 1 to 3 years are highly sensitive to changes in environmental temperature, especially during the summer season. Excess heat can disturb their sleep, cause discomfort, and lead to crying or irritation. Maintaining a comfortable temperature is an important

aspect of child care. According to pediatric doctors, a surrounding temperature of around 28°C is considered safe and comfortable for young children.

This project, titled “Temperature Controlled Child Care System for Children Using Arduino UNO”, is designed to monitor the surrounding temperature and provide an audio response when the temperature exceeds the safe limit. The system uses an LM35 temperature sensor to continuously sense the ambient temperature near the child. An Arduino UNO processes the sensor data and compares it with a preset threshold value.

When the temperature crosses 28°C, a DFPlayer Mini audio module activates a speaker to play a soothing song, indicating that the child may be uncomfortable and needs attention. If the temperature remains within the safe range, the system stays silent. The proposed system is simple, cost-effective, and suitable for basic child-care support applications.

Backend Theory

The backend of this project is based on sensing and control using Arduino UNO. The LM35 temperature sensor converts ambient temperature into an analog voltage, which is read by the Arduino. The Arduino compares this value with a preset threshold of 28°C. When the temperature exceeds the limit, Arduino sends serial commands to the DFPlayer Mini, which plays an audio signal through a speaker. The system works on real-time monitoring and simple decision-based control.

Methodology Used

The methodology of the Temperature Controlled Child Care System Using Arduino UNO involves real-time temperature monitoring and audio-based care indication. The process begins with powering the Arduino UNO using a 5V supply. The LM35 temperature sensor continuously senses the surrounding temperature near the child and converts it into an analog voltage proportional to the temperature.

This analog signal is fed to the Arduino UNO through an analog input pin. The Arduino processes the sensor data by converting it into a digital value and calculating the corresponding temperature in degrees Celsius. The measured temperature is then compared with a predefined threshold value of 28°C, which is recommended by pediatric doctors as a comfortable temperature for children aged 1 to 3 years.

If the temperature is less than or equal to 28°C, the system remains inactive and no sound is played. When the temperature exceeds 28°C, the Arduino sends serial commands to the DFPlayer Mini audio module. The DFPlayer Mini accesses the audio file stored in the microSD card and plays a soothing song through the connected speaker. This indicates that the environment has become uncomfortable and child care attention is required. The system continuously monitors temperature and responds automatically, ensuring simple and effective child care support.

Components required

Table no, 1 Component details

Sl. No	Component Name	Specification / Description	Quantity
1	Arduino UNO	ATmega328P microcontroller board	1

2	Temperature Sensor	LM35 (Analog temperature sensor)	1
3	Audio Module	DFPlayer Mini MP3 module	1
4	Speaker	4Ω / 8Ω mini speaker	1
5	MicroSD Card	≤32 GB, FAT32 format	1
6	Resistor	1 kΩ (for DFPlayer RX protection)	1
7	Jumper Wires	Male-to-male / male-to-female As required	8
8	Breadboard	For temporary connections	1
9	Power Supply	5V (USB or adapter)	1
10	Connecting Cables	USB cable for Arduino	1

Expected results:

The proposed Temperature Controlled Child Care System is expected to continuously monitor the surrounding temperature of children aged 1 to 3 years. When the temperature remains at or below 28°C, the system stays inactive and no audio is played. When the temperature rises above 28°C, the Arduino UNO successfully detects the change and activates the DFPlayer Mini module. A soothing song is played through

the speaker to indicate that the child may feel discomfort due to excess heat. The system operates reliably in real time, especially during the summer season, and helps caregivers identify uncomfortable temperature conditions easily.

Pin-diagram:

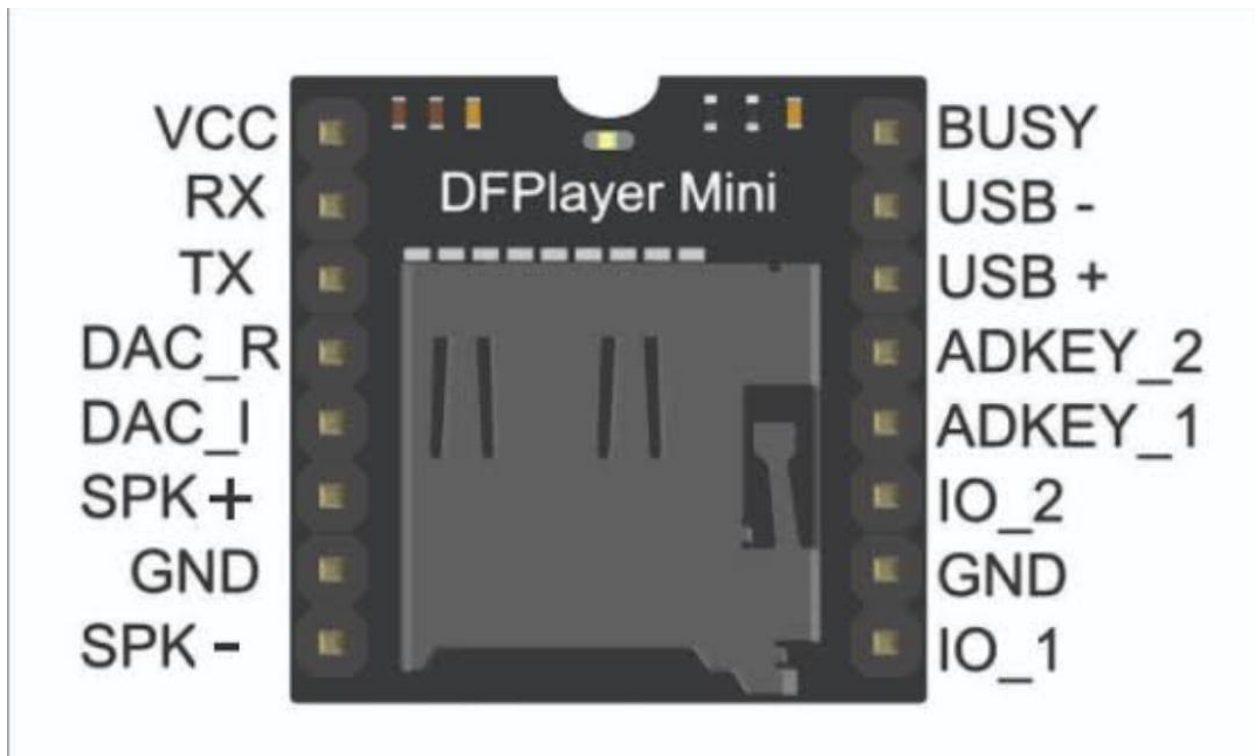


Fig. no.1 MINI DF PLAYER

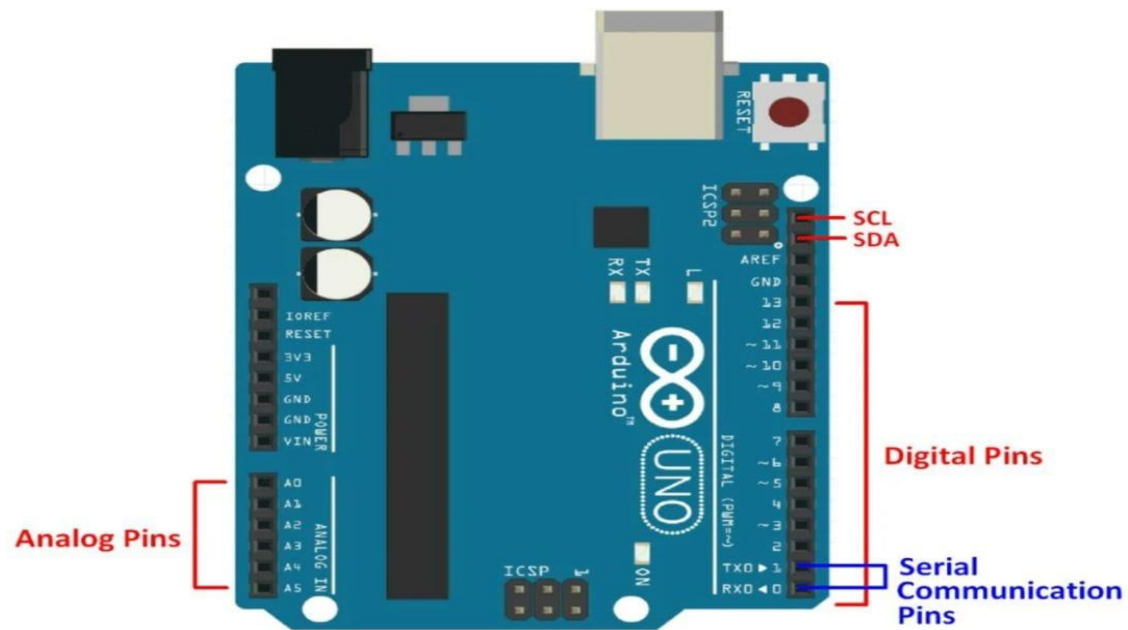


Fig. no. 2 ARDUINO UNO

Circuit diagram:

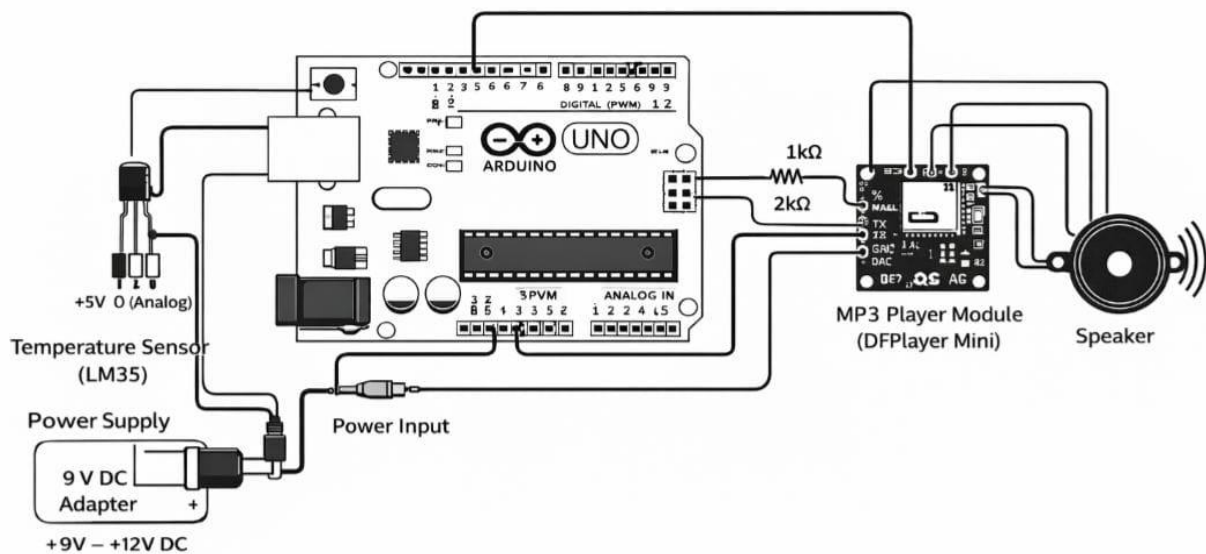


fig. no. 3 main daigram

CODE USED FOR ARDIUNO UNO :

```
#include <SoftwareSerial.h>

#include <DFRobotDFPlayerMini.h>

#define LM35_PIN A0

SoftwareSerial mySerial(10, 11); // RX, TX

DFRobotDFPlayerMini myDFPlayer;

bool songPlaying = false;

void setup()
{
  Serial.begin(9600);
  mySerial.begin(9600);
  if (!myDFPlayer.begin(mySerial)) {
    Serial.println("DFPlayer not detected");
    while (true);
  }
  myDFPlayer.volume(0); // start with mute
}

void loop() {
  int sensorValue = analogRead(LM35_PIN);
  float voltage = sensorValue * (5.0 / 1023.0);
  float temperature = voltage * 100;

  Serial.print("Temperature: ");
  Serial.println(temperature);

  // ♦ If temperature crosses 28°C
  if (temperature > 28) {
    if (!songPlaying) {
      myDFPlayer.play(1); // play 0001.mp3
      songPlaying = true;
    }
    // 🔊 Slowly increase volume (fade-in)
```

```

for (int vol = 0; vol <= 25; vol++) {
    myDFPlayer.volume(vol);
    delay(150); // controls speed of slow start
} } } else {
    // ♦ If temperature less than or equal to 28°C
    if (songPlaying) {
        myDFPlayer.stop();
        myDFPlayer.volume(0);
        songPlaying = false; }}
    delay(1000);}

```

Progress mode:

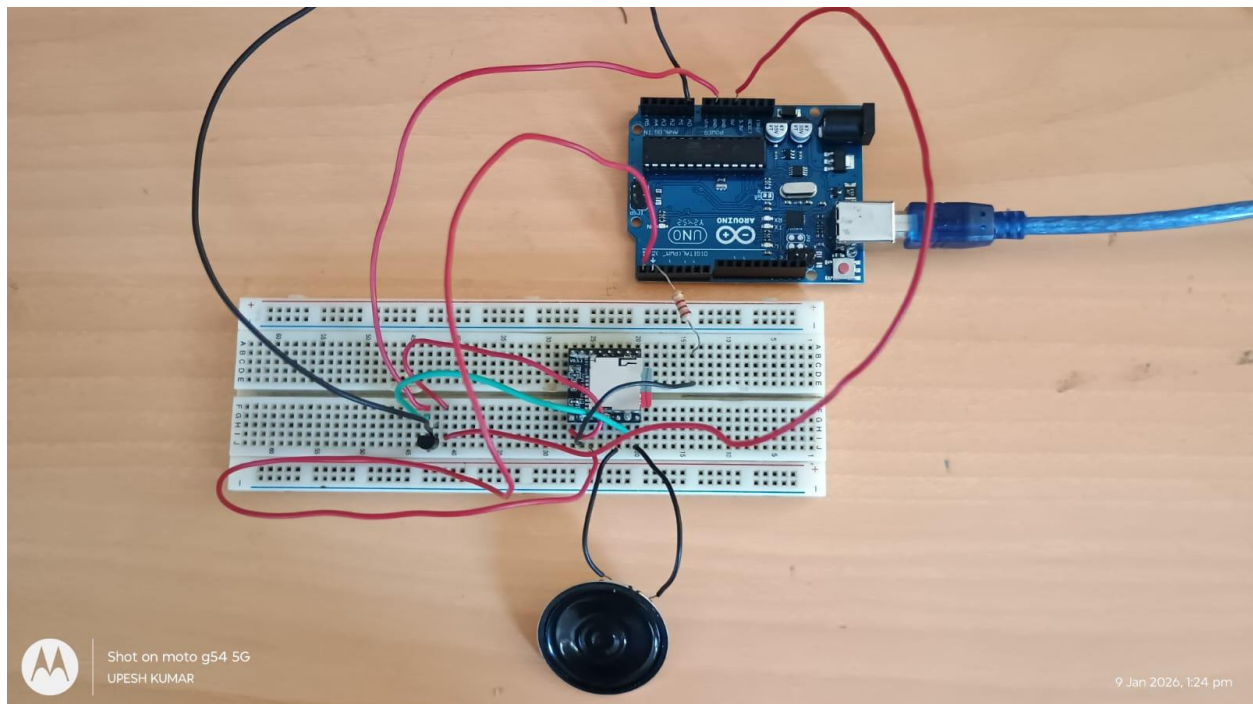


fig. no . 4

Applications

- Used for children aged 1 to 3 years to maintain a comfortable environment during summer.
- Automatically plays soothing music when temperature exceeds 28°C, helping to calm children.
- Reduces present-day mobile phone addiction in babies by avoiding the use of mobile songs.
- Provides music without screens, preventing early exposure to mobile radiation and screen dependency.
- Can be safely used in smart toys and musical toys for children.
- Helps parents avoid giving mobiles to babies for calming or entertainment purposes.
- Safer alternative to mobile phones and other toxic electronic gadgets.
- Suitable for homes, day-care centers, and child-care environments.
- Low-cost, low-power, and child-safe electronic system.
- Encourages healthy child development using dedicated hardware-based care.

References:

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